

Catalytic cracking

Timestamped narration script | 58-second video

Read each line during its marked interval. Aim for about 140 words per minute. Pronounce FCC as F-C-C and LPG as L-P-G.

TIME	ON SCREEN	SPOKEN SCRIPT
00:00 - 00:04	FCC overview	Fluid catalytic cracking converts heavy gas oil into lighter products.
00:04 - 00:11	Feed and catalyst	Preheated oil meets hot catalyst in the riser. Steam disperses the feed, while catalyst heat vaporises it.
00:11 - 00:16	Riser flow	Vapour and catalyst rise together. Zeolite acid sites help break carbon-carbon bonds.
00:16 - 00:20	Cracking and coke	Cracking absorbs heat. Coke deposits gradually reduce catalyst activity.
00:20 - 00:24	Cyclone separation	Cyclones separate catalyst rapidly. Product vapour enters the fractionator.
00:24 - 00:28	Steam stripping	Steam strips retained hydrocarbons before the catalyst enters the regenerator.
00:28 - 00:33	Coke combustion	Air burns coke, releasing heat and restoring catalyst activity.
00:33 - 00:38	Catalyst return	Hot catalyst returns to the riser. Flue gas leaves for cleaning.
00:38 - 00:43	Fractionation	The fractionator separates products by boiling range. Overhead streams enter condensation and gas recovery.
00:43 - 00:48	Product recovery	Gas recovery yields LPG and dry gas. Gasoline, cycle oils and slurry also leave.
00:48 - 00:54	Heat balance	Circulating catalyst carries combustion heat back to vaporisation and cracking.
00:54 - 00:58	Closing loop	Crack, separate, regenerate, repeat: continuous conversion with continuous catalyst regeneration.

Matched to Catalytic_Cracking_Explainer_BlendedLogo_58s.mp4. The video is silent; these lines are for live delivery or a separate voice-over.